

NIKITA BAND

Full-Stack Developer | AI-Integrated Systems
+91-7028162051 | [Email](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

PDPM IIITDM Jabalpur

B.Tech in Electronics & Communication Engineering

Jabalpur, MP
2023 – Present

Jawahar Navodaya Vidyalaya, Akola

Class XII: 91% | Class X: 98%

Maharashtra

DSA & Problem Solving

- Solved 300+ LeetCode problems (DP, Graphs, Trees, Binary Search, Sliding Window), Rating 1671 on LeetCode | 2 ★ CodeChef, Rating 1510

Experience

Full-Stack MERN Developer Intern — Wallpull Solution Pvt. Ltd.

Remote | Nov 2025 – Jan 2026

- Architected 6+ reusable React component libraries with Context API-driven state management, cutting prop-drilling by ~60% across 3 sustainability research dashboards
- Integrated 12 RESTful API endpoints enabling real-time data sync; reduced client-server round-trips by 35% through request batching and debounced calls
- Engineered Node.js/Express middleware pipeline with JWT authentication, role-based access control, and input validation — eliminating a critical auth bypass vulnerability
- Refactored 8 class components to React Hooks + memoization, reducing unnecessary re-renders by ~40% and improving Lighthouse performance score from 64 → 89
- Shipped 4 full-stack features end-to-end in 2-week sprints collaborating with a cross-functional team of 5 engineers using Git-based code review workflows

Projects

DevFlow CI | Personal Project | [Live Demo](#) | [GitHub](#)

- Built a full-stack AI-powered code review platform that automatically analyzes GitHub pull requests and posts inline review comments, using Node.js, TypeScript, React, PostgreSQL, and Redis
- Architected an asynchronous, queue-based processing pipeline with Redis and BullMQ to decouple webhook ingestion from AI analysis, returning gateway acknowledgments in under 50ms while reviews process independently with retry logic and a dead-letter queue
- Integrated Google's Gemini 2.5 Flash to generate structured, severity-tagged code review comments, parsing AI output into GitHub's PR review API format via Octokit and a GitHub App bot
- Implemented secure authentication with GitHub OAuth, JWT-based Bearer tokens, AES-256 encrypted credential storage, and HMAC-SHA256 webhook signature verification with Redis-based atomic idempotency to prevent duplicate event processing
- Deployed as three independently running services (React dashboard, API gateway, background worker) on Render with managed PostgreSQL and Redis

LinkUp | Personal Project | [Live Demo](#) | [GitHub](#)

- Built a production-grade real-time video conferencing platform from scratch using WebRTC for peer-to-peer video and Socket.IO for signaling, with ICE candidate queuing for restricted network reliability
- Integrated Sarvam AI (saaras:v3) for live Hindi-English speech translation, including RMS energy-based voice activity detection to eliminate hallucinated transcriptions on silence
- Implemented real-time ML hand gesture detection using MediaPipe (21 hand landmarks), triggering synchronized particle effects across all participants via Socket.IO broadcast
- Built AI-generated meeting summaries using Gemini, plus screen recording, speaking indicators via AudioContext FFT analysis, and live captioning
- Deployed frontend on Vercel and backend on Render with MongoDB Atlas for persistence

StockTradingPlatform | Personal Project | [Live Demo](#) | [GitHub](#)

- Built a full-stack Zerodha-style trading platform end-to-end using React, Node.js/Express, and MongoDB Atlas to understand real trading system architecture
- Implemented JWT authentication with dynamic username extraction from token payload, plus Holdings, Positions, Orders, and Funds dashboard views with live P&L calculations
- Integrated Alpha Vantage API for live Indian stock prices with 10-minute caching to manage free-tier rate limits
- Deployed across three separate AWS CloudFront distributions (frontend, dashboard, backend proxy) backed by S3 static hosting and an EC2 instance running Node.js via PM2
- Wrote 10 Jest/Supertest integration tests covering all API endpoints

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, Java, C++, SQL

Frontend: React.js, Tailwind CSS, HTML, CSS, Context API, React Hooks, Responsive UI, + Vite, React Router v6, Axios, shadcn/ui

Backend: Node.js, Express.js, REST APIs, JWT Authentication, WebSockets (Socket.io), Rate Limiting, Scalable API Design, GitHub OAuth 2.0, Webhook Handling & HMAC-SHA256 Verification, Asynchronous Job Queues (BullMQ), Dead-Letter Queue Pattern, Microservices Architecture, Express Middleware (Helmet.js, CORS), Idempotent Request Handling

Databases: MongoDB, SQL, Schema Design, Aggregation Pipelines, PostgreSQL, Prisma ORM, Redis

AI / ML: LLM Integration (LLaMA), Prompt Engineering, NLP Basics, Machine Learning, Gemini API, Prompt Injection Defense, Structured JSON Output Parsing

Testing & Tools: Jest, Postman, Git, GitHub, AWS (EC2, S3), Vercel, CI/CD, Render

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, System Design

Security: AES-256 Encryption, HMAC-SHA256 Signature Verification, HTTP Security Headers, CORS Configuration, OAuth Token Management

Achievements

- Amazon Future Engineer Scholar (AFE) • Finalist – BitByBit Hackathon, Cognizance'26, IIT Roorkee
- Programming in Java – NPTEL (IIT Kharagpur), Elite (90%), 2025
- NVS Regional Topper – Class X (98%) • All India Rank 515 – B.Arch JEE 2022